Visit **www.brpaper.com** for downloading previous years question papers of
10th and 12th (PSEB and CBSE), IKPTU,MRSSTU, PSBTE, PANJAB UNIVERSITY,
PUNJABI UNIVERSITY, BFUHS, HPTU, HPSBTE , HARYANA DIPLOMA, MDU HARYANA

Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

ADVANCED STRUCTURAL ANALYSIS

Subject Code : PECE-603C-18

M.Code : 79402

Date of Examination: 19-12-22

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

1. Write briefly :

a) Distinguish between static indeterminacy and kinematic indeterminacy.

b) What do you mean by equilibrium and compatibility?

c) Define the term rotation factor used in Kani's method.

d) Draw a neat sketch of discretization of a continuum.

e) Out of displacement method and force method, which method is better for the analysis of rigid jointed plane frames and why?

f) Derive Flexibility matrix for the truss element.

g) Describe about structural similitude.

h) Write the assumptions which are involved in portal method of analysis.

i) Define action and displacement.

j) Mention the limitations of FEM.

SECTION-B

2. Describe the stiffness matrix of elements in global coordinates from element coordinates.

3. Using flexibility matrix method, analyze the beam shown in fig. 1.

BR paper
exams preparatory

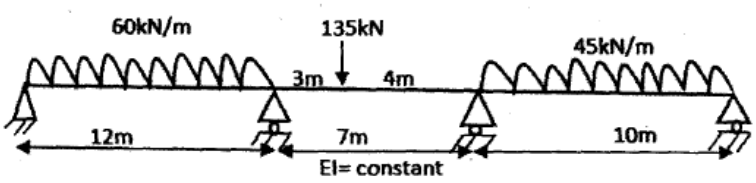

Fig.1

4. Discuss various steps of finite element method for analyzing a structure.

5. Define the term rotation factor used in the Kani's method and derive the expression for it.

6. '*Structures may be solved by either stiffness matrix method or flexibility method. Generally, stiffness matrix is preferred'*. Explain.

SECTION-C

7. Analyse the continuous beam, shown in fig. 2 using Stiffness Method and develop the BMD.

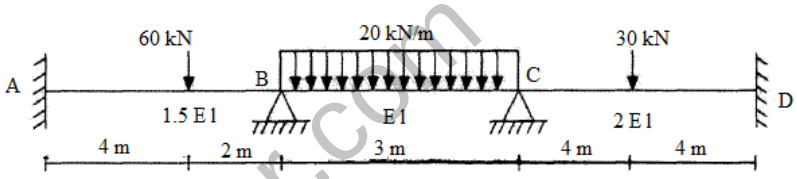

Fig. 2

8. Analyze the frame shown in fig.3, by flexibility matrix method.

   Ll = 6 m, L2 = 3.5 m, L3 = 6 m, And P = 40 kN; w =25kN/m

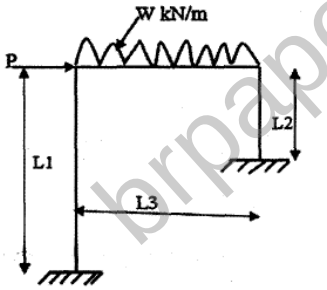

Fig.3

9. Write short note on following :

   a) Concept of physical approach.

   b) Write short note on substitute frames.

brpaper.com
brpaper.com